



Experimental study on small group behavior and crowd dynamics in a tall office building evacuation

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HIGHLIGHTS

- Experiments considering different social relations are conducted in a tall office building.
- Space-time features, speed characteristics and density-speed relations for each experiment are analyzed.
- Speed discrepancies for each pair and impacts of small groups on flow are discussed.
- Small groups make positive or negative impacts on crowd dynamics with depending on the competitive mechanism.

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ABSTRACT

It is well known that a large percentage of occupants in a building are evacuated together with their friends, families, and officemates, especially in China. Small group behaviors are therefore critical for crowd movement. This paper aims to study the crowd dynamic considering different social relations and the impacts of small groups on crowd dynamics in emergency evacuation. Three experiments are conducted in an 11-storey office building. In the first two experiments, all participants are classmates and know each other well. They are evacuated as individuals or pairs. In the third experiment, social relations among the participants are complex. Participants consist of 8 families, 6 lovers and several individuals. Space-time features, speed characteristics and density-speed relations for each experiment are analyzed and compared. Results conclude that small group behaviors can make positive impacts on crowd dynamics when evacuees know each other and are cooperative. This conclusion is also testified by four verified experiments. In the third experiment, speeds of evacuees are lowest. Small groups form automatically with the presence of intimate social relations. Small groups in this experiment slow down the average speed of the crowd and make disturbance on the crowd flow. Small groups in this case make negative impacts on the movement of the crowd. It is because that evacuees do not know each other and they are competitive to each other. Characteristics of different types of small groups are also investigated. Experimental data can provide foundational parameters for evacuation model development and are helpful for building designers.

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1. Introduction

Collective behavior and evacuation dynamics have been investigated for several decades. As we have known, crowd motion is driven by the interaction between occupants and surrounding during evacuation. Crowd consists of not only

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individuals but also occupants in groups [1]. Research has found that up to 70% of people in the crowd are actually moving in group under normal condition [2]. A group exists in form of lovers, friends, families and so on, which often have two to four members. Cohesion between groups' members is so strong that members intend to stay together even in emergency [3]. It is quite common in office and residential buildings as well as shopping malls during evacuation. For example, according to survey, more than 85% people form groups with others during the "9.11" WTC disaster and most of them come from the same or adjacent office [4]. Obviously, the relationship may influence the behavior of the members of group, such as competition and cooperation behaviors during evacuation [5]. Furthermore, the groups may also affect people around them since the groups may become obstacle for the surrounding people during evacuation and finally affect the pattern and movement of the whole crowd [6,7]. It can be concluded that the presence of different social relations and various types of the groups may make a big difference to the characteristics of crowd dynamics.

Pedestrian group behaviors have been investigated by the experts in different research areas, such as, sociology, physics, and mathematics. Experiments and simulations are both performed. Experimental studies mainly focus on the groups' movement characteristics [2,8,9] and affected factors on them [10–12] in normal condition as well as decision-making behaviors [13]. Very limited studies focus on small groups' behaviors, moving speeds, and their impacts on crowd flow in emergency. Simulation studies focus on developing or improving pedestrian models by considering group effect. Social force model [2,14,15], field floor model [16,17] and agent-based model [18–22] are among widely employed models. They have been modified and extended to study group behaviors and their impacts on crowd movements. Even though simulation researches have carried out on this topic for relative simple scenarios in normal and in emergency, the parameters used in the model should be verified and analyzed. Experiments considering various groups in different scenarios should be investigated.

Stair, as an essential exit in the building, plays an important role in emergency evacuation. Evacuation using the stairs is more complex than evacuation in the rooms or corridors. It is significant to study evacuation in staircase. Crowd dynamic parameters and factors which affect evacuation efficiency and human behaviors have been discussed. Fruin et al. [23], Proulx [24], Peacock et al. [25,26], Fang et al. [27], and Kuligowski [28] studied movement characteristics of pedestrians and concluded that a series of movement speed ranges were possible in the different staircase evacuation scenarios. These data provided the foundation for future code development. They also suggested that motion characteristics of evacuees depend on many factors, like gender, age. Song, Huo and Ma et al. [29,30] particularly investigated the speed and time features along with transit behaviors in an high-rise building evacuation and it was found that overtaking phenomena were observed frequently on the refuge floor. Fujiyama [31] developed a model to predict the walking speed of pedestrians in the staircase. Parameters such as building layout, physical conditions of the occupants, and numbers of the exits all make significant impacts on evacuation efficiency. Pauls et al. [32] explored the minimum stair width for evacuation. A series of width are listed for different types of stairs. Gwynne et al. [33] and Qu et al. [34] simulated the movement of pedestrian on the stairs. They pointed out that staircase geometries such as the number of steps and the depth of treads affect the movement of the evacuees. Lee et al. [35] conducted evacuation experiments under different widths of corridors and stairs. It was concluded that evacuation behaviors were mainly influenced by the locations of the stairs. Lei et al. [36] observed a dormitory evacuation and used FDS+Evac software to discuss the width of the stair and the exit on flow rates, it was founded the existence of stairs caused flow stratification, and the larger the exit width, the earlier the stratification phenomenon appeared. Galea et al. [37] and Boyce et al. [38], investigated how various factors, such as stair widths, population characteristics and geometrical locations of the floor, influenced merging behavior. Ding et al. [39] simulated stair evacuation with the improved cellular automata model, evacuees' walk preference behavior is considered in the model. Yang et al. observed [40] small groups behaviors during staircase evacuation, but only their impacts on crowd flow are mentioned. Even though staircase evacuation has been investigated widely, social relations and small group behaviors are barely considered.

This paper aims to study the crowd dynamic considering different social relations and the impacts of small groups on crowd dynamics in the staircase experiments. Crowds consist of individuals, pairs and a mixture of individuals and small groups respectively. The article is organized as follows. Section 2 introduces the experiment setup and method. Sections 3 and 4 presents the detailed results of experiments and discussion. Lastly Section 5 is the summary of our work.

2. Experimental setup and method

2.1. Building layout

Controlled experiments in our study are conducted in an 11-storey office building in Tsinghua University in China. All evacuation experiments start on the tenth floor in the building, whose layout is presented in Fig. 1. There are two staircases and two elevators in the building. However, only Stair 1 is used in the experiments. The structure and dimension of Stair 1 is shown in Fig. 2. It is 1.2 m wide while each step is 0.15 m high and 0.3 m deep. Step numbers in each floor are different. There are 25 steps in each floor from floor three to floor ten while 35 steps in floor two. It is noted that floor one is the ground floor. The travel distance between each floor can be formulized as follows [27]:

$$L = n_s l_{\text{inclination}} + (n_f - 1) l_{\text{turning}}, \quad (1)$$

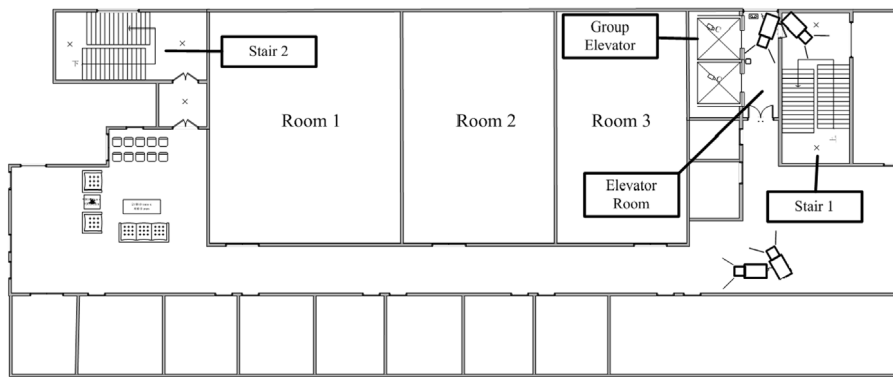


Fig. 1. Layout of the tenth floor of the office building.

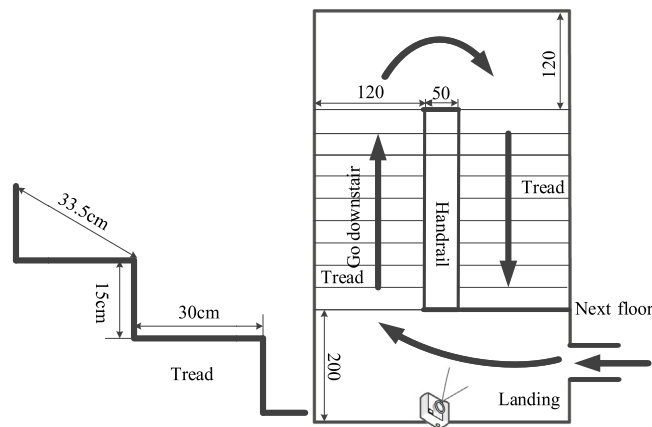


Fig. 2. Geographic structure and dimension of the stair 1.

where $l_{inclination}$ is the inclination length, which depends on the depth of tread and the height of the step, and the value is 0.335 m in our study. $l_{turning}$ donates the turning length, which indicates the movement distance of an evacuee on a landing when turning from one stair flight to another, and the value is 1.18 m in this paper. The step number in each floor is n_s , as mentioned above its value is 35 for floor two and 25 for others. n_f represents the number of landing in each floor section, whose value is 4 for floor ten and two and 3 for others.

To collect the data of these experiments, eight cameras are deployed at different places to record the process of each evacuation. Four of them are placed on the top of the stair landings of the 9th, 7th, 5th and 3rd floors, two are located on the top of elevator lobby on the 1st and the 10th floors, and other two are deployed at the right side of the corridor of the 10th floor, as shown in Fig. 1. To track the participants and deal with the data precisely, a hat with number is requested to wear for each evacuee in the experiments. Due to these cameras, arrival time and departure time along with the behaviors of each evacuee on different floors can be measured. Fig. 3 shows a snapshot of an experiment on the fifth floor.

2.2. Experimental procedure

The objective of this study is to investigate the impacts of small group and social relation behaviors on crowd dynamics in staircase evacuation. Three evacuation experiments are carried out in the building, as shown in Table 1. 30 undergraduate students composed of 11 boys and 19 girls from Renmin University of China take part in the first and second experiments, and they are all 18–22 years old. They all come from the same class and know each other very well. Evacuation scenarios are different for the first and second experiments. In the first experiment, all participants are evacuated as individuals while in the second experiment, every two persons are required to form a pair and all participants are evacuated in pairs. It is noted that every pair is formed in advance according to the questionnaire results, which reflect their willing, the elaborate method has been introduced in another paper [41]. In the third experiment, there are 56 participants and 27 of them are college postgraduate students, who contain 7 pairs of lovers and several individuals. Among the participants, 23 of them come from



Fig. 3. Snapshot of an experiment.

Table 1
Scenario of each experiment.

Experiment	Participants number	Egress exit	Scenario
1	30	Stair 1	In individual
2	30	Stair 1	In pair
3	56 (8 families, 6 lovers)	Stair 1	Individual, subgroups

the same institute and know each other while other 4 are girlfriends of four participants in the staircase. Other 29 are local residents consisting of 8 groups of families and some individuals. They are unfamiliar with the others. Meanwhile, among all participants in the experiment, 18 of them are female and 8 of them are children under 14 years old.

Before each evacuation experiment, all participants are distributed randomly in Room 1. All evacuees run to the elevator lobby and choose stair 1 as the exit to move down once the experiment starts. They succeed only they arrive to the destination on the ground floor. To enhance the participants' motivation and make experiments more realistic, a reward measure is taken up: the top three who are evacuated successfully will be awarded. When the participants are evacuated in pairs, "the top three" means top three pairs who must arrive on the first floor together.

3. Results

3.1. Speed analysis

In three experiments, all participants are evacuated from the 10th floor to the first floor. In order to understand each experiment and to investigate the movement characteristics of the participants, space–time features and average speeds in different floor sections of each participant for three experiments are presented in Figs. 4–6. Since the third experiment is much different from the first two experiments, the first two experiments are analyzed firstly. As shown in Figs. 4(a) and 5(a), it is found that total evacuation time for two experiments are 119 s and 114 s, respectively. Space–time features in the first experiment differ from those in the second one. Compared with the first experiment, lines in the second experiment are more incompact, which means relatively greater time internal discrepancy among the participants in the second experiment. Overtaking phenomenon occurs in both experiments. Such phenomenon is relatively rare compared with other studies [30] but it is more apparent in the second experiment than in the first one. The possible reason is that the participants come from the same class and have collaborative behaviors. They communicate with others and do not intend to overtake others when evacuated together in the first experiments. However in the second experiment the participants may consider more for the partners. The members in each pair will wait for each other. When one falls behind, the other member will slow down. The participants around him/her will overtake him/her. Figs. 4(b) and 5(b) exhibit the speeds of each evacuee in different floor sections. The method used to calculate the speed of each evacuee is elaborated by Fangs' study [27], which can be expressed by Eq. (2). V donates the speed of each evacuee. l is the travel distance and ΔT stands for the time each evacuee spends moving during each or two adjacent floor. It can be seen that almost all participants move downward at more or less same speed. Very few participants change their speeds in the experiments. However, in the second experiment, the first two participants entering the staircase in F10-9 section, F9-7 section and F7-3 section travel faster than the others, and then they reduce the speed to approximately same one with the others. It can be also observed that the members in pairs wait for

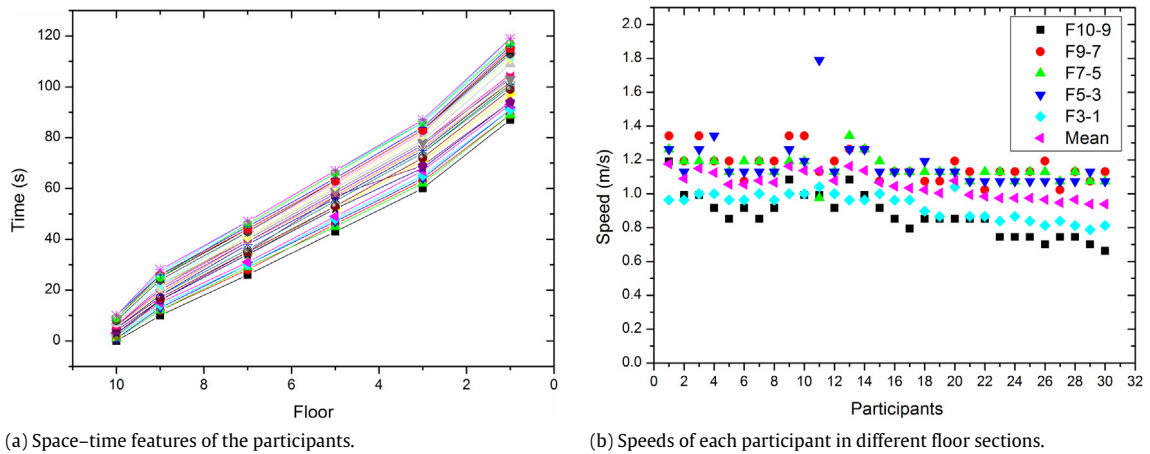


Fig. 4. Space-time features and speeds of each participant in the first experiment.

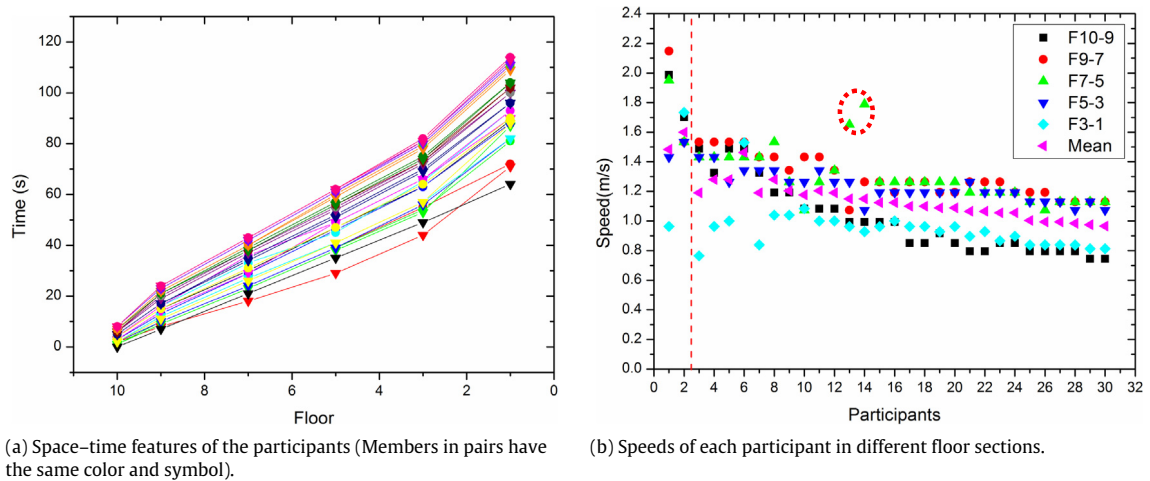


Fig. 5. Space-time features and speeds for each participant in the second experiment.

each other and adjust their speeds simultaneously. For instance, No. 13 and No. 14 accelerate their speeds simultaneously in F7-5 section, denoted by the dotted circles in Fig. 5(b).

$$V = \frac{l}{\Delta T}. \quad (2)$$

Fig. 6(a) demonstrates space-time features of the participants in the third experiment. In order to analyze space-time features of various types of participants, lines are plotted with three colors. Black, red and green lines denote male, female and children participants, respectively. From the figure, it can be found that total evacuation time is 201 s. Some participants have the same space-time features, which suggests they move together all the time. Combining video data with the figure, it can be found most participants follow others to move downward and male participants prefer to overtake others comparing with female participants in the experiment. It is noted that three male participants do not respond to the rule and run to the stair 2 at the beginning. They are far behind the others at the first observation area, as the shown in the solid circle in Fig. 6(a). But they catch up with the others at the fifth floor and blend in with the crowd. Children behave quite differently. The speed for the child traveling alone is faster than that for the child traveling with the parents. It can also be found that multi-layer phenomenon becomes more obvious as the decrease of the floor. The crowd is clustered into two sub-groups on the seventh floor and three sub-groups on the first floor.

Fig. 6(b) shows the speeds of each participant in different floor sections. The speed of each participant is a little distinct at the same floor especially for the child traveling alone and the male participant during F10-5 section who does not respond to the rule firstly, as shown in the dotted and solid circle respectively, it is also obvious for the speed during F3-1 section. Fig. 7 presents average speeds of male, female and children participants. It can be found that average speeds of children participants are greater than others. Average speeds of male participants are also higher than those of female evacuees. It is

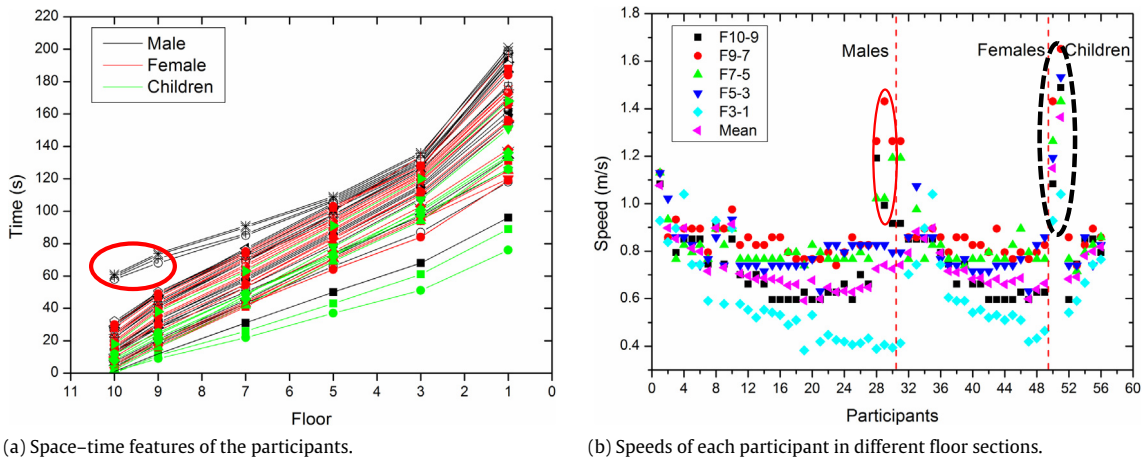


Fig. 6. Space-time features and speeds of each participant in the third experiment. (The dotted and solid circles display the speeds of children traveling alone and the male participants during F10-5 section who do not respond to the rule firstly, respectively.)

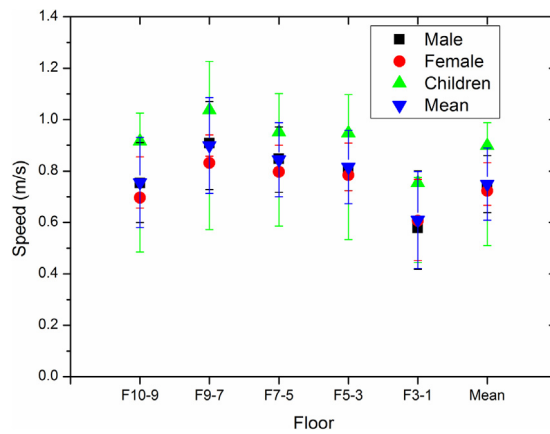


Fig. 7. Average speeds of each type of the participants.

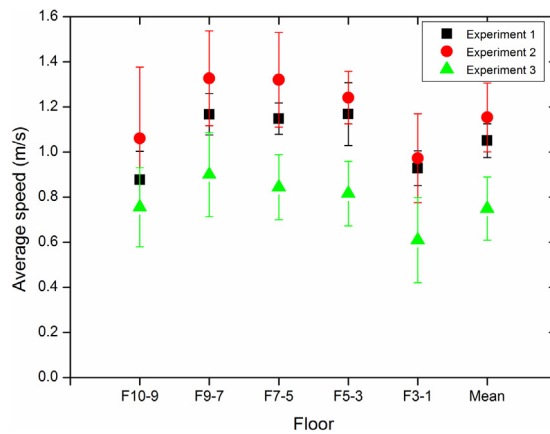


Fig. 8. Average speeds of all participants in different floor sections.

quite surprised that children run fast in the experiment. It may be because that they are more serious to the award and run fast in the experiment.

Fig. 8 demonstrates the average speeds of the participants in different floor sections for each experiment. It is clearly shown that average speeds of the participants in different floor sections have the same tendency as the decrease of the floor for each experiment. The speed increases firstly and reaches to the peak at F9-7 section, and then remains the relatively

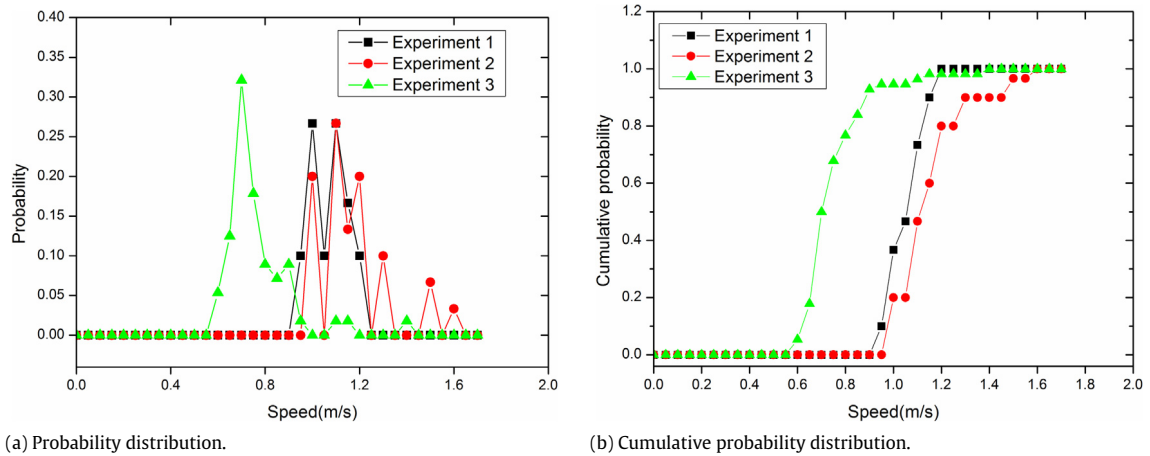


Fig. 9. Distribution of average speeds for the first, second and third experiments.

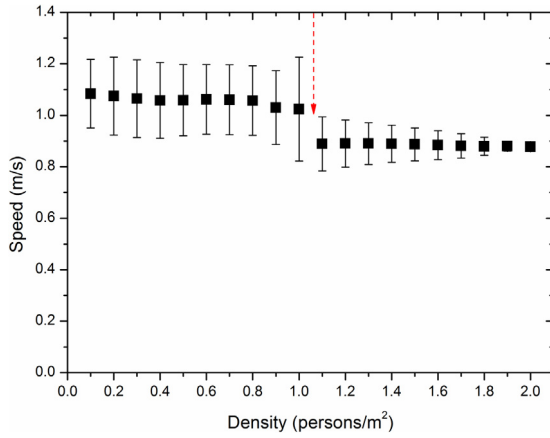
stable value in F7-5 and F5-3 sections. Lastly, in F3-1 section, the speed drops down to a lower value. The tendency is reasonable. It is explained that the participants are clustered together at the stair door on the tenth floor and entered the staircase with the lower speeds to learn the surroundings as well as to ensure safety in F10-9 section. They will then increase the speeds and keep similar speeds to the destination as soon as possible. Finally, they slow down due to fatigue and move through the stair exit on the first floor. The overall average speeds for the participants in three experiments are 1.05 ± 0.07 m/s, 1.15 ± 0.15 m/s and 0.75 ± 0.14 m/s, respectively. Comparing average speeds in each floor section for the first two experiments, it is easily found that average speeds as well as their standard deviations in the second experiment are higher than those in the first experiment. When the participants are evacuated as individuals, they have the almost same physical condition and similar descent velocity. The standard deviation is therefore very small. In the second experiment, the members in pairs show cooperative behaviors. The faster runner in the pair will intend to help the slower runner. Also, none of the slower runner would like to hold the partner back. It is hard for pairs to become obstacles in the second experiment because pairs also collaborate with each other expect for cooperative behaviors between members in each pair. The average speeds in the second experiment are higher. On the other hand, great speed discrepancies among pairs occur. The possible explanations are the speed discrepancies among pairs are larger. For instances, pairs with fast members run even faster, which results in largest speed discrepancy with the lowest pair. Social relation may be beneficial for evacuation, which is owned to effective cooperative behaviors among evacuees.

Speeds in the third experiment are the lowest. Unlike the participants in the first two experiments, social relations among the participants in the third experiment are complex, which contain classmates, families and lovers. Observing the video data carefully, it is found that most families and lovers move down together and form 11 small groups during evacuation. Small groups especially for families protect each other and are unfamiliar with the others. They do not intend to cooperate with the others and also they do not want to give way to the others, which reflects individuals or small groups are competitive to each other. Therefore, groups are easy to become obstacles and slow down others' speeds. The overall speed of the crowd depends on the slowest evacuees. As a result, the overall speed may become smaller. Results suggest that social relation and small groups may make adverse impacts on evacuation when evacuees do not know each other.

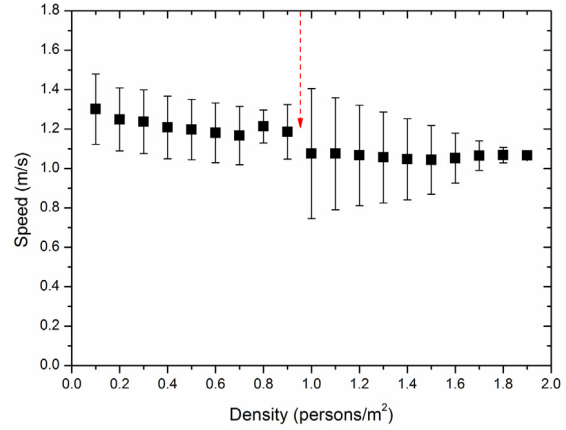
Fig. 9 shows the distributions of overall average speeds of evacuees. The maximum speeds for three experiments are 1.18 m/s, 1.60 m/s and 1.37 m/s, respectively. Fig. 9(a) presents the probability distribution of average speeds, in which each point denotes the probability of corresponding speed for evacuees. There are two peaks in the curve for the first experiment, which suggest evacuees whose speeds between 0.96 and 1 m/s and also between 1.06 and 1.1 m/s account for the most of all. In the second experiment, evacuees who run between 1.05 and 1.1 m/s take up 27%, the most of all. A peak is therefore resulted. In the third experiment, there is an evident peak in the curve, which indicates that evacuees moving at the speed between 0.71 and 0.75 m/s account for the largest proportion of the all. Fig. 9(b) demonstrates the cumulative probability distributions of the speeds. Analyzing the curves, it can be found that all curves show similar shape even though the range of the speeds is quite different. Only 10% evacuees move slower than 1 m/s in the first experiment while all evacuees move faster than 1 m/s in the second experiment. However, in the third experiment, only about 5% evacuees move faster than 1 m/s.

3.2. Fundamental diagrams

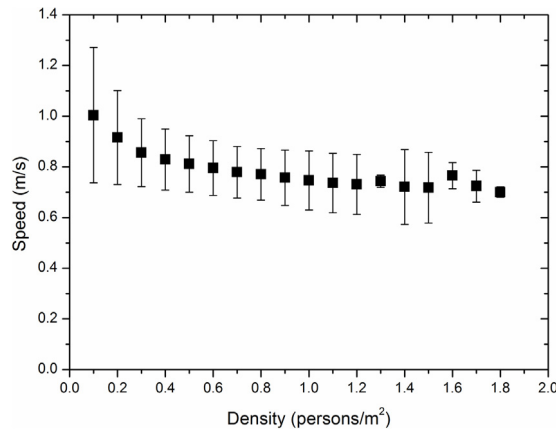
Fig. 10 shows the relations between average density and speed for three experiments. The method to obtain the data is elaborated by Huo' thesis [42]. $\rho(n\Delta t) = \frac{\Delta N(n\Delta t)}{d \cdot l}$ and $\bar{v}(n\Delta t) = \frac{\sum_{i=1}^{\Delta N'(n\Delta t)} v(i)}{\Delta N'(n\Delta t)}$ are used to calculate the density and speed at the $n\Delta t$ in each or two adjacent floor section, where d is the width of the stair, l represents the travel distance of evacuees



(a) Fundamental diagram in the first experiment.



(b) Fundamental diagram in the second experiment.



(c) Fundamental diagram in the third experiment.

Fig. 10. Fundamental diagram of evacuees in three experiments.

during measured floor section and $\Delta N(n\Delta t)$ donates the number of evacuees at the $n\Delta t$ during measured floor section. $\Delta N'(n\Delta t)$ contains not only the number of evacuees at the $n\Delta t$ but also the evacuees leaving the measured floor section at or before $n\Delta t$ but going through the area between $(n-1)\Delta t$ and $n\Delta t$. $v(i)$ stands for the speed of each evacuee. Based on the calculation method of density and speed, it is easy to plot the relations of density and speed for each or two adjacent floor. To make the foundational diagram more distinct, we make further statistics to the average densities and average speeds for the whole staircase. The densities and speeds in Fig. 10 donate the statistics of the average densities and average speeds for the whole staircase. It can be observed that the speed decreases with the density, which are consistent with results of previous researches [27,30]. Comparing Fig. 10(a) and (b), it is obvious that the speed in the second experiment is higher than that in the first experiment for the same density. It is noted that the speed declines significantly when the density is at 1.1 person/m² in the first experiment and the density is at 1.0 person/m² in the second experiment. The reason is that there is no sufficient space for evacuees to speed up when the density is at 1.1 person/m² in the first experiment and the density is at 1.0 person/m² in the second experiment. The speed of evacuees in the third experiment is the lowest comparing with other two experiments. Meanwhile, the tendency of speed with density in the third experiment is also a little different from other two experiments. The speed falls down smoothly and there is no sudden change with the decrease of the density.

3.3. Characteristic of small groups

In Section 3.1, it is discussed average speeds of the participants evacuated in pairs are higher than those evacuated as individuals. To further understand speed difference in two experiments, the average speeds of each pair are compared, as shown in Fig. 11. It is obvious that each pair's speed in the second experiment is higher than that in the first experiment. A parameter ε_{cg} is defined to donate the degree of speed discrepancy for each pair, which can be expressed as follows,

$$\varepsilon_{cg} = \frac{v_2 - v_1}{v_1} \quad (3)$$

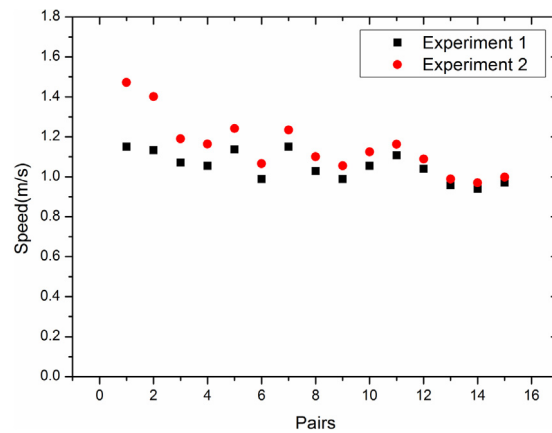


Fig. 11. Comparison of the speeds for each pair.

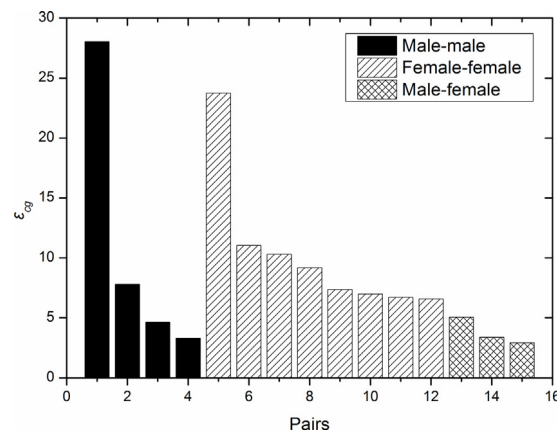


Fig. 12. Comparison of the speeds for each pair.

where v_1 and v_2 represent the average speeds of a pair in two experiments, respectively. Fig. 12 presents the degree of speed discrepancy for different types of the pairs. It can be found the speed discrepancy for the pairs consisting of the members with the same gender are higher than those for the pairs made of members with different genders. Meanwhile comparing to male–male pairs, speed improvement for most female–female pairs are more obvious. One possible explanation is that the members with the same gender in a pair have similar speeds and cooperate better in evacuation than the members with different genders in a pair. Female members prefer to move together in emergency and can cooperate well with the others. From this perspective, female–female pairs may cooperate better than male–male pairs.

As mentioned before, there are 11 small groups in the third experiment. Comparing to individuals, the members in small groups may move differently and make impacts on the movement of crowd. Table 2 elaborates the members and their social relationships, in which M, F, B and G signify male, female, boy and girl participants. There are two pairs of mother and daughter, a pair of father and daughter, a couple, six pairs of lovers and a family of three members. Fig. 13 presents the speed of each member in 11 groups. As shown in the figure, it can be seen that the members in each group intend to move downward with a consistent speed. Parents protect their children and keep their speeds consistent with their children. The family, couple and lovers collaborate with each other. It is also observed that the speed discrepancies of groups 1, 2, 3, 6 and 7 are smaller than those of groups 4, 5, 8, 9, 10 and 11 in different floor sections. The reason is that parents always adjust their speeds and move discreetly to ensure the safety of children. Such groups move down at a stable speed. Group 6 to group 7 have better and similar athletic abilities and they run in front with few obstacles. They therefore change their speeds little during evacuation.

To investigate impacts of small groups on the movement of the crowd, Fig. 14 compares the average speeds of small groups and the crowd in different floor sections. It can be found that the average speeds of small groups show the same tendency with the crowd. However, the average speeds of small group are always smaller than those of the crowd. The average speed of the crowd from floor ten to floor one is 0.75 m/s while the average speed of small groups is 0.73 m/s. It suggests that small groups in the crowd slow down the average speed of the crowd. Fig. 15 presents the average flow of the participants on the third floor landing for the third experiment, which is useful to further reveal impacts of small groups on crowd movement. The Equation used to calculate the flow is $F = \frac{\Delta N}{\Delta T}$, where F and d donate the flow and the width

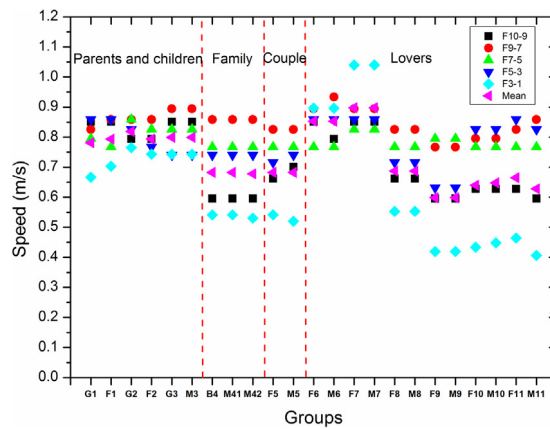


Fig. 13. Speeds of each member in each pair.

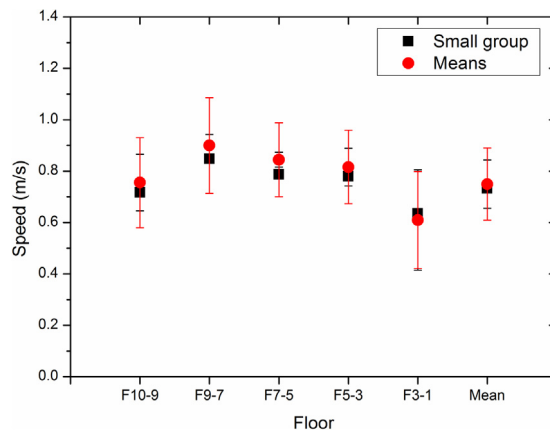


Fig. 14. Comparison of speeds for small groups and the crowd.

Table 2

Eleven small groups in the third experiment.

Groups	1	2	3	4	5	6	7	8	9	10	11
Relationship	Mother and daughter		Father and daughter		Family	Couple		Lover			
Members	G1,F1	G2,F2	G3,M3	B4,M41,M42		F5,M5	F6,M6	F7,M7	F8,M8	F9,M9	F10,M10 F11,M11

Table 3

Speed and influence of the first person or couple on others' movement at the flow peaks.

Peak No.	1	2	3	4	5	6	7
First person/couple	Boy	Boy	Man	Woman	Mother and daughter	Couple	Lover
Speed	1.53 m/s	1.19 m/s	1.13 m/s	1.07 m/s	0.85 m/s	0.71 m/s	0.85 m/s
Influence	Lead	Lead	Lead	Lead	Obstruct	Obstruct	Lead

of the stair respectively. ΔT is the time interval, it equals to 3 s, which is a constant here. ΔN represents the number of participants arriving at the stair landing during each ΔT . There are seven peaks at the flow curve. Table 3 shows who is the first person or couple arriving at the landing at the flow peaks and his/their speed as well as his/their influence on other evacuees behind him/them. It is noted that a mother–daughter pair and a couple arrive at the third floor landing at 96 s and 200 s respectively. Their speeds are lower than people before them and they obstruct the evacuees after them. Therefore, congestion and flow peaks appear. Conversely, a pair of lovers cooperates well and they take leadership for the evacuees after them. Their speed is higher than people before them and drive the evacuees after them to move fast, and new flow peaks also appear. In general, small groups make disturbance on the crowd flow and may bring congestion and risk to the crowd.

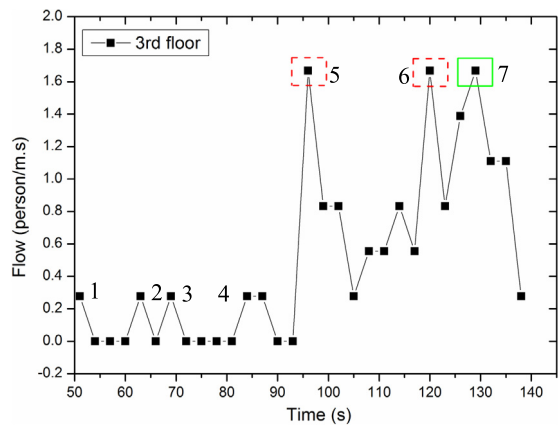


Fig. 15. Flow against time at the 3rd floor in the third experiment.

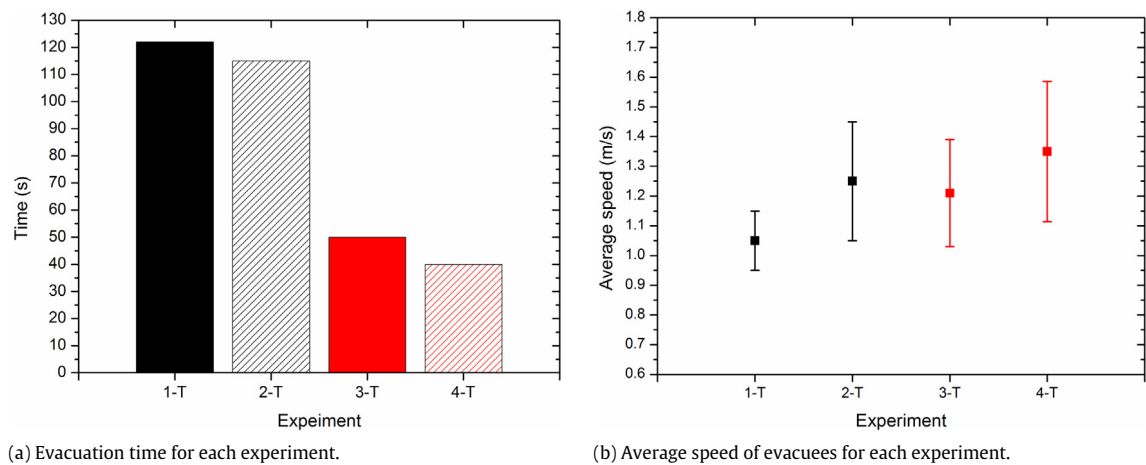


Fig. 16. Evacuation time and average speed of evacuees for each verified experiment.

Table 4
Scenario of four verified experiments.

Experiment	Egress exit	Floor	Scenario
1-T	Stair 1	10-1	In individual
2-T	Stair 1	10-1	In pair
3-T	Stair 1	10-7	In individual
4-T	Stair 1	10-7	In pair

4. Discussion

In three experiments, speeds in each floor section agree well with previous data from various experiments [25,26]. However, overall average speeds in staircase during our experiments, 1.05 ± 0.07 m/s for individuals, 1.15 ± 0.15 m/s for pairs and 0.75 ± 0.14 m/s for complex groups, are quite different from the data obtained by Peacock [25] in another 10-layer building. This may be because small group behaviors and social relations among evacuees are not considered sufficiently in previous study. Meanwhile, other factors, such as age, gender and shape of evacuees, population density and stair geometries may also affect the movement of the evacuees.

To testify the repeatability of the experiments and conclusion, other four experiments were also conducted in the same office building, as shown in Table 4. There are 21 participants in these four experiments and they all come from the same participants of the first experiment mentioned above. In experiment 1-T and experiment 2-T, all participants are evacuated from the tenth floor to the first floor as individuals and in pair respectively. In experiment 3-T and experiment 4-T, all participants are evacuated from the tenth floor to the first floor as individuals and in pair respectively.

Fig. 16 shows the evacuation time and average speeds of evacuees for each verified experiment. It is obvious participants evacuated in pair take less time and run averagely faster than participants evacuated as individuals. It suggests collaboration

behavior results in fast evacuation. Small group behaviors can make positive impacts on crowd dynamics when evacuees know each other and are cooperative.

In most previous studies, results revealed that small group behaviors resulted in adverse effects on pedestrian movement [14–16]. However, simulation results made by Vizzari et al. [22] suggested that, in low to moderate density ($0\text{--}2.5$ person/m²), the presence of groups brought negative effect while it made positive influence on crowd flow at high density ($2.5\text{--}4$ person/m²). In this work, experimental results differ from previous studies. Compared average speeds and evacuation time in the first two experiments and four verified experiments, it is obvious that when evacuees know each other and they evacuate together with whom they prefer to, evacuation efficiency is improved. The third experiment is quite different from the first two experiments, not all evacuees know each other and social relations among them are complex. It is found that small groups form automatically with the presence of intimate social relations during evacuation and they are competitive to others. Speeds of evacuees are lowest and small groups slow down the average speed of crowd, as discussed in Section 3.3, which also suggest small groups bring adverse impact on the movement of evacuees. When analyzing the relation between speed and density, it is concluded the density of evacuees in floor sections ranges $0\text{ person/m}^2\text{--}2.2\text{ person/m}^2$ for all three experiments. The speeds of evacuees are lowest in the third experiment and the speeds are the highest in the second experiment. Result is different from the results of Vizzari et al. [19]. All results show that the effects of small groups on crowd are complex, social relations among evacuees need to be considered.

5. Conclusions

This paper aims to study the impacts of small groups and social relations on crowd dynamics in the staircase during evacuation. Three controlled evacuation experiments are conducted in an 11-storey office building. In the first and second experiments, all participants are classmates and know each other well. They are evacuated as individuals or pairs. In the third experiment, social relations among the participants are complex and not all of them know each other. Participants consist of 8 families, 6 lovers and several individuals. Firstly, space–time features of evacuation process and speeds of evacuees for each experiment are analyzed and compared. It is found that overall average speeds for three experiments are 1.05 ± 0.07 m/s, 1.15 ± 0.15 m/s and 0.75 ± 0.14 m/s, respectively. Experiment results suggest that evacuation efficiency is improved when evacuees know each other and move down together as pairs. The conclusion is also testified by four verified experiments. Our conclusion is contradictory to the previous studies. However, it agrees with practice that evacuation efficiency may be improved if the slowest moving persons are helped by others. Speeds in the third experiment are the lowest. Small groups block others and slow down the average speeds of crowd if small groups do not know each other well. It means that the impacts of small groups on crowd dynamics are strongly related to competition and cooperation behaviors among the evacuees. Secondly, the relation between speed and density is studied. It is concluded that the speed of evacuees decreases with the density for all experiments. At the same density, the speed of evacuees is the lowest in the third experiment and highest in the second experiment. The speed of evacuees declines significantly when the density is at 1.1 person/m^2 in the first experiment and the density is at 1.0 person/m^2 in the second experiment. Lastly, characteristics of small groups are discussed. The speed discrepancies for each pair or each small group and their impacts on flow are analyzed. It is found that small groups in the third experiment slow down the average speed of crowd and bring congestion to the crowd.

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